

# Soil Doctor : Embedded AI-Based Soil Analysis Device for Crop and Fertilizer Recommendation

Miniproject Group 05

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# Introduction

- **Soilnutrient imbalance** directly affects crop growth, yield, and overall soil health.
- **Traditional soil testing** involves manual sampling + lab analysis, causing an “**information lag**” that delays timely intervention.
- Reliance on **experience-based guesswork** often leads to improper fertilizer usage, resulting in soil degradation and reduced yields.
- **Our solution:** an ESP32-based portable device for real-time measurement of NPK levels and instant crop & fertilizer recommendations.
- **An industrial 7-in-1 multifunctional sensor** is used to capture NPK, pH, EC, and moisture data in real time.
- Integrates **Embedded AI** by deploying a **trained Decision Tree model** on the ESP32, enabling offline inference without cloud dependency.
- Provides up to 95% accurate, real-time agricultural insights and on-site recommendations, even in remote areas without **internet access**.

# Motivation

| Societal Need                                                                                                      | Technical Need                                                                                              |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Empowering Smallholder Farmers:</b> Affordable, handheld alternative to expensive and infrequent lab testing    | <b>Eliminating Information Lag:</b> Replaces delayed lab analysis with instant, real-time field results     |
| <b>Maximizing Agricultural Output:</b> Data-driven insights instead of guesswork to improve crop quality and yield | <b>Edge AI for Remote Areas:</b> Decision Tree on ESP32 enables offline inference without internet          |
| <b>Resource Efficiency:</b> Optimizes fertilizer usage for better soil health and reduced environmental impact     | <b>Industrial-Grade Precision:</b> 7-in-1 sensor captures NPK, pH, EC, and temperature in one portable unit |

# Literature Survey

| Title & Publication                                                                                                                                                  | Author                                                                     | Methodology / Key Findings                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI-Driven Smart Agriculture: An Integrated Approach for Soil Analysis, Irrigation, and Crop-Fertilizer Recommendations (2025, IEEE Access - Research Article)</b> | A. Khaliq, A. Khan, S. Jan , M. Umair, A. Gulshair, A. Ali, and U. A. Shah | <ul style="list-style-type: none"><li>- The paper highlights the use of a 7-in-1 sensor to collect essential real-time field data, specifically measuring soil moisture, temperature, pH, and macronutrient levels (Nitrogen, Phosphorus, and Potassium).</li><li>- A core theme of the paper is moving beyond traditional "fixed rules" or human intuition to provide data-driven crop and fertilizer recommendations using Machine Learning.</li></ul> |

# Literature Survey

| Title & Publication                                                                                                                                                                          | Author                                                                           | Methodology / Key Findings                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prediction of Crop Yield Based-on Soil Moisture using Machine Learning Algorithms(2022,2nd International Conference on Technological Advancements in Computational Sciences (ICTACS))</b> | S.Madhuri,S Paudel,R Nakarmi,P Giri, S B Karki                                   | -The paper specifically identifies the Decision Tree algorithm as a highly effective method for data-driven decision-making in agriculture, noting that it achieved a 95% accuracy rate in predicting crop yields. |
| <b>Tamil Nadu Agricultural University (TNAU)(2024,Agritech Portal)</b>                                                                                                                       | Online : Available: <a href="http://agritech.tnau.ac.in">agritech.tnau.ac.in</a> | -Data Collection for crops.<br>-Key attributes : Nitrogen (N),Phosphorus (P),Potassium (K),pH,Electrical Conductivity (EC),Temperature (°C).                                                                       |

# Problem Formulation

## Gap in Existing Systems :

- Information Lag: Traditional soil testing (manual + lab-based) is time-consuming and expensive, making frequent monitoring impractical
- Reliance on Guesswork: Lack of real-time data leads to improper fertilizer application
- Environmental & Economic Impact: Inaccurate usage causes soil degradation, environmental damage, and higher costs
- Connectivity Dependency: Many smart systems require internet, limiting use in remote rural areas

## Problem Statement

Design and implement a portable ESP32-based soil monitoring system to measure soil nutrient (NPK) levels and provide appropriate crops and fertilizer recommendations for enhanced agricultural productivity and soil health.

# Problem Formulation

## Project Objectives :

- Real-Time Monitoring: Develop a portable ESP32-based device to measure NPK, pH, EC, moisture, and temperature using a 7-in-1 sensor
- Embedded Decision Making: Implement a Decision Tree model for offline crop and fertilizer recommendations
- Precision Agriculture: Provide a low-cost, edge-based solution to improve productivity and ensure sustainable soil health

# Functional Requirements

- 7-in-1 Real-Time Sensing: Measures N, P, K, pH, EC, moisture, and temperature simultaneously
- Real-Time Processing: ESP32 acquires and analyzes soil data instantly
- Embedded AI Inference: Decision Tree-based crop recommendation executed offline on ESP32
- Rule-Based Fertilizer Advice: Identifies nutrient deficiencies using agronomic thresholds (TNAU-based)
- Hardcoded ML Logic: Python-trained model implemented as C if-then rules for edge inference
- User Interface: Displays soil data, crop, and fertilizer suggestions on 16×2 LCD with button control
- Portable Operation: Battery-powered, standalone system with no internet dependency

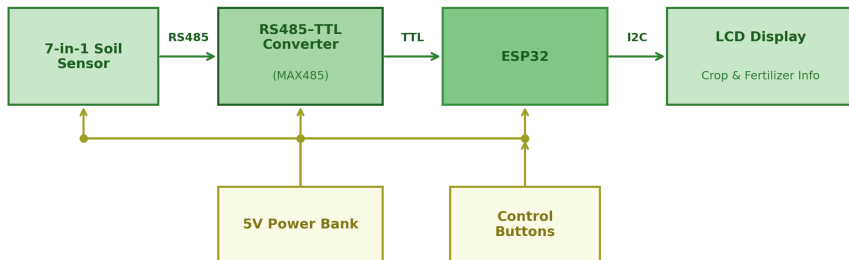
# Module Description

The system is divided into five primary modules, each playing a critical role in transforming raw soil data into actionable agricultural intelligence at the field's edge.

- **Sensor Interface Module** : Industrial **7-in-1 multifunctional sensor** for real-time acquisition of Nitrogen, Phosphorus, Potassium, pH, EC, Moisture, and Temperature.
- **Microcontroller Unit (MCU) Module** : **ESP32 Dev Board**; serves as the central brain for data processing, signal parsing, and AI rule execution.
- **Communication Module** : **MAX485 converter**; bridges RS485 sensor signals with ESP32 TTL logic to ensure stable data transmission in field conditions.
- **Software & AI Module** : Embedded **Decision Tree logic** providing offline, 95% accurate crop and fertilizer recommendations via hard-coded "if-then" branches.
- **User Interface (UI) & Power Module** : **16x2 I2C LCD** and push-buttons for local results visualization, powered by a **5V rechargeable unit** for standalone portability.

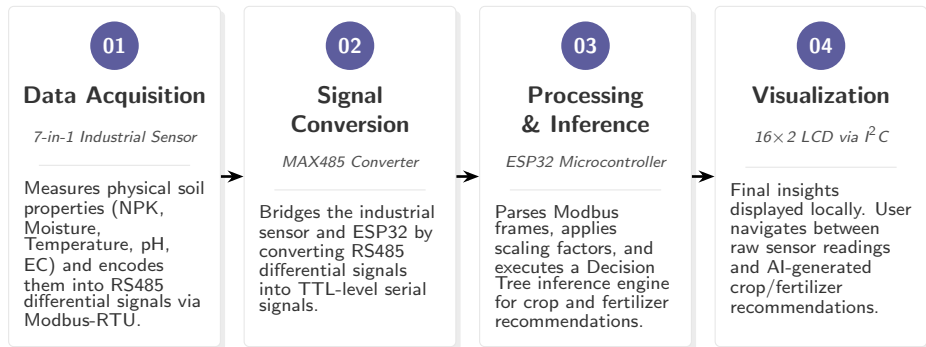
# Block Diagram

## Embedded Soil Analysis System for Crop & Fertilizer Recommendation



# Data Flow Stages

## Four Stage Processing Pipeline



Sequential pipeline: each stage feeds the next, ensuring data integrity from soil to insight

# Signal Paths & Control Logic

## Half-Duplex RS485 Control

GPIO 4 → RE/DE

- Dedicated GPIO 4 toggles RE/DE pin.
- **TX Mode:** ESP32 asserts DE HIGH.
- **RX Mode:** ESP32 pulls RE LOW.
- Requires strict direction switching.

## Modbus-RTU Protocol

4800 Baud | CRC-16

- Serial communication at 4800 baud.
- 16-bit Cyclic Redundancy Check.
- CRC mismatch rejects the frame.
- Ensures integrity in field conditions.

## User Navigation Logic

GPIO 18 → Button

- GPIO 18 with internal pull-ups.
- Software debounce logic included.
- Cycles display modes on each press.
- **Modes:** NPK → Environment → AI.

# Data Transformations

## Three Levels of Intelligence

### Level 1 Frame Trans- formation

High Byte + Low Byte  
→ 16-bit Integer

Raw hexadecimal bytes from the sensor's Modbus response are reconstructed into 16-bit unsigned integers by combining high and low bytes from each register.

### Level 2 Numerical Scaling

Raw  $\div$  10.0  
→ Real-World Value

Scaling factors from the sensor datasheet are applied. Moisture, Temperature, and pH raw values are divided by 10.0 to match physical resolutions.

### Level 3 Categorical Mapping (AI)

Moisture  $\leq$  57.5%  
→ "Cowpea"

Numeric data is passed through a Decision Tree inference engine derived from agronomic benchmarks, mapping inputs to categorical crop outputs.

**Raw bytes → Scaled floats → Actionable agricultural intelligence**

# Implementation Details

## Hardware Tools

| Component                                                        | Description                                                                                |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| ESP32 Dev board                                                  | Main microcontroller for data processing and control                                       |
| Five-pin soil multi-parameter sensor (ZTS-3002-TR-ECTHNPKPH-N01) | Measures 7 soil parameters: temperature, moisture, EC, pH, nitrogen, phosphorus, potassium |
| RS485–TTL Converter (MAX485)                                     | Converts RS485 sensor signals to TTL for ESP32 communication                               |
| 16×2 LCD with $I^2C$                                             | Displays soil parameters and recommendations                                               |
| Push Button                                                      | Mode selection & user navigation                                                           |
| Power Switch                                                     | ON/OFF control of the system                                                               |
| Wires & Passive Components                                       | Electrical connections and signal conditioning                                             |
| Power Bank                                                       | Portable custom-built 5V rechargeable power supply                                         |

# Implementation Details

## Software Tools

| Software Tool          | Description                                                                                          |
|------------------------|------------------------------------------------------------------------------------------------------|
| EasyEDA(Std)           | For electronic schematic capture and designing the pinout mapping.                                   |
| Spyder IDE             | Utilized to train, test, and extract rules from the Decision Tree crop recommendation model.(Python) |
| Wokwi Online Simulator | Employed for functional verification of the firmware and circuit logic before physical assembly      |
| Arduino IDE            | The primary environment for firmware development, debugging, and ESP32 flashing.                     |

# Implementation Details

## Datasets/Protocols

| Datasets/Protocols                           | Description                                                                                                                                      |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Modbus-RTU (over RS485)                      | Communication Protocol, utilizing a standard 4800 baud rate and 16-bit Cyclic Redundancy Check (CRC) for industrial-grade data integrity.        |
| TNAU<br>(Tamil Nadu Agricultural University) | Agronomic Dataset, which provide the optimal nutrient thresholds for crops.                                                                      |
| AI Rules                                     | A rule-based "if-then" inference engine derived from a Decision Tree trained on moisture and nutrient ranges to provide offline recommendations. |

## External APIs/Interfaces

To maintain full functionality in remote rural areas, the system is designed for offline, edge-based inference and does not require external cloud APIs for its core operations. The primary hardware interface is the RS485 serial link used to communicate with the industrial soil sensor.

# Team Task Allocation

The project responsibilities were distributed to leverage the specific strengths of each team member while ensuring collaborative effort for the final implementation stages.

- Circuit Design: Megha C and Gopika Suresh
- Software Development: Midhunrag N V and Shreya S N
- PCB Layout & Fabrication: Jointly handled by all team members
- Hardware Assembly & Soldering: Jointly handled by all team members
- System Testing & Calibration: Jointly handled by all team members
- Documentation & Reporting: Jointly handled by all team members

# Project Timeline and Milestones

| SI No | Project Phase                                                                             | Date of Completion |
|-------|-------------------------------------------------------------------------------------------|--------------------|
| 1     | Problem Identification and Literature Survey                                              | January 15, 2026   |
| 2     | System Design and Component Selection                                                     | January 29, 2026   |
| 3     | Circuit Design and Algorithm Development                                                  | February 7, 2026   |
| 4     | Decision Model of Fertilizer Recommendation,Simulation and Functional Verification(Wokwi) | February 12, 2026  |
| 5     | Hardware Assembly and Sensor Interfacing                                                  | February 20, 2026  |
| 6     | System Integration and Initial Testing                                                    | February 27, 2026  |
| 7     | Sensor Calibration & Minor Improvements                                                   | March 5, 2026      |
| 8     | PCB Design & Fabrication                                                                  | March 12, 2026     |
| 9     | Results Analysis                                                                          | March 15, 2026     |
| 10    | Final Demonstration                                                                       | March 30, 2026     |

## Circuit Diagram

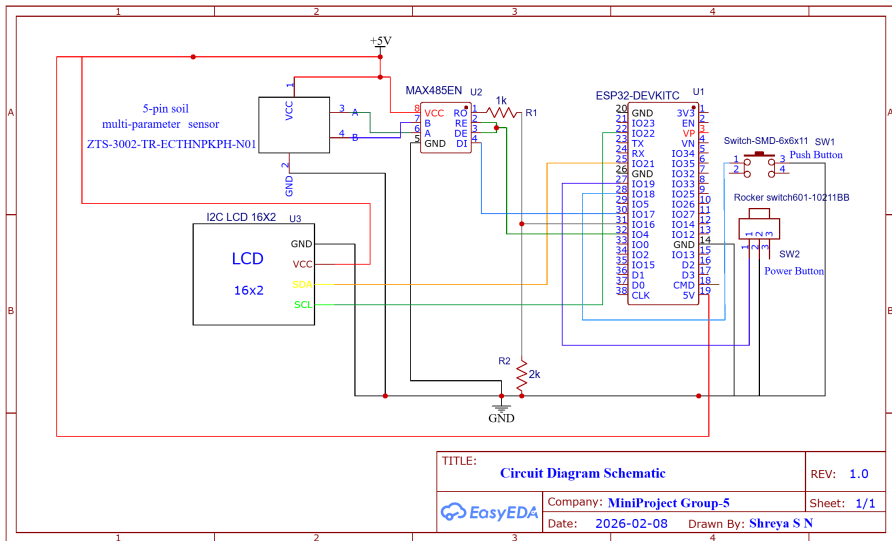


Fig. 2. Circuit Diagram

# Results - Hardware Setup & Implementation



Fig. 3. Fully integrated portable system in its transparent enclosure

# Results - Simulation Outputs

## Functional Simulation:

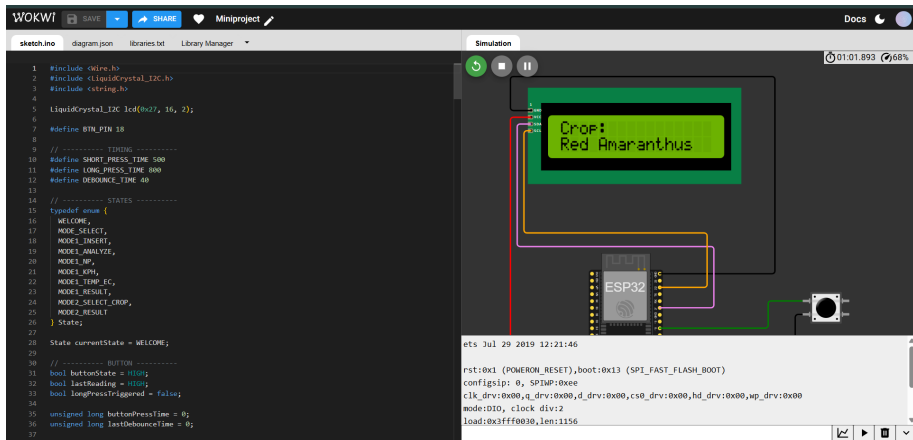


Fig. 4. Wokwi simulation showing the virtual LCD successfully parsing and displaying emulated soil data

# Results - Machine Learning Model (Decision Tree)

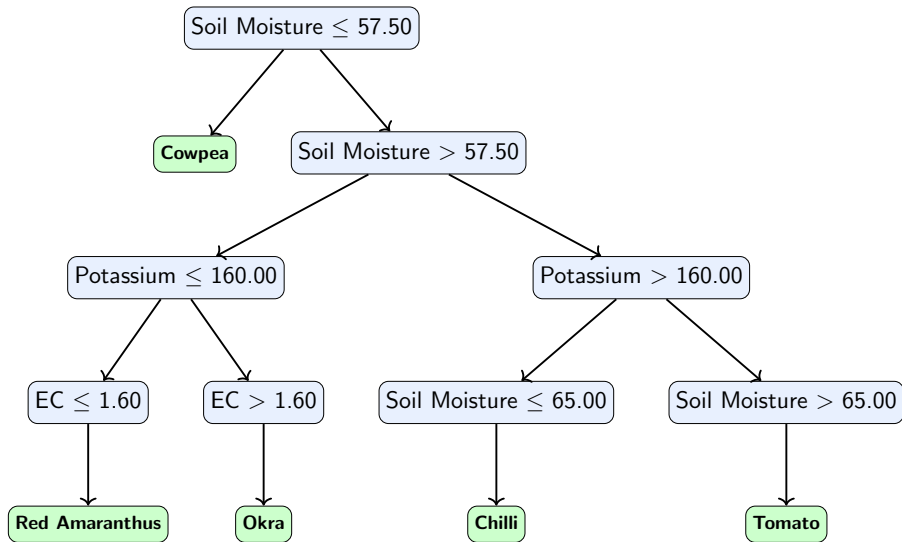
- A **Decision Tree Classifier** is used for crop recommendation due to its interpretability.
- Trained on soil parameters—N, P, K, pH, Temperature, EC, and Soil Moisture—the model generates clear, human-readable decision rules.
- These rules are directly implemented on the **ESP32**, enabling accurate offline crop recommendation without internet dependency.

## DECISION TREE RULES:

```
|--- Soil Moisture <= 57.50
|   |--- class: Cowpea
|--- Soil Moisture > 57.50
|   |--- Potassium <= 160.00
|       |--- EC <= 1.60
|           |--- class: Red Amaranthus
|           |--- EC > 1.60
|               |--- class: Okra
|       |--- Potassium > 160.00
|           |--- Soil Moisture <= 65.00
|               |--- class: Chilli
|           |--- Soil Moisture > 65.00
|               |--- class: Tomato
```

Fig. 5. Decision Tree Rules for Crop Classification

# Results - Machine Learning Model(Decision Tree Classifier)



# Results - Functional Requirements

Real-Time Data Acquisition: Photos of the LCD displaying live parameters:



Fig. 6. Cowpea

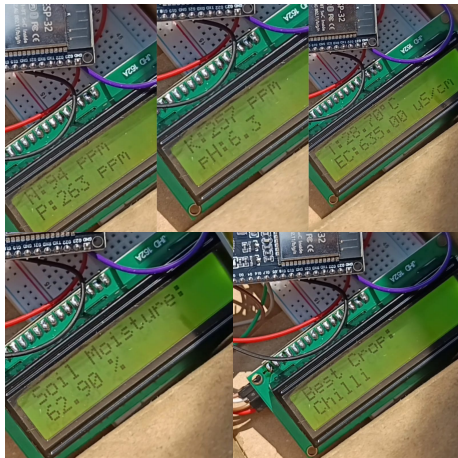


Fig. 7. Chilli

# Results - Functional Requirements



Fig. 8. Figure showing High Nitrogen



Fig. 9. Fertilizer Logic Output

# Results - Validation Table

## Test Case Validation

The system was tested with different sample NPK values to verify crop classification and fertilizer recommendation logic. The observed outputs are summarized below:

Soil Parameter Test Cases and Crop Recommendations

| TC | N<br>(ppm) | P<br>(ppm) | K<br>(ppm) | pH<br>Range | Temp<br>(°C) | Moist.<br>(%) | EC<br>(uS/cm) | Rec.<br>Crop   | Fertilizer<br>Suggestion |
|----|------------|------------|------------|-------------|--------------|---------------|---------------|----------------|--------------------------|
| 1  | 563        | 1347       | 1350       | 6.9         | 28.6         | 65.7          | 2883          | Tomato         | Apply Urea               |
| 2  | 298        | 736        | 733        | 7.1         | 28.4         | 51.3          | 1615          | Cowpea         | Apply Phosphorus         |
| 3  | 7          | 62         | 54         | 6.4         | 29.6         | 58.2          | 218           | Red Amaranthus | Apply Potassium          |
| 4  | 218        | 551        | 548        | 9           | 28.6         | 61.5          | 1226          | Chilli         | Reduce N fert            |

# Evaluation

The system's performance was evaluated by comparing the design objectives (Expected) against the functional test results (Actual) across five key technical metrics.

| Metric             | Expected Performance                               | Actual Performance                                                            |
|--------------------|----------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Latency</b>     | Real-time response to eliminate "Information Lag." | <b>&lt;10 Seconds</b> ; quick offline inference via ESP32 logic.              |
| <b>Power</b>       | Low-power, portable for long-term field use.       | Stable operation via 5V rechargeable bank ; fully regulated standalone power. |
| <b>Reliability</b> | Noise-resistant transfer in harsh environments.    | Error-free via MAX485 bridge and 16-bit <b>CRC</b> to prevent corruption.     |
| <b>Throughput</b>  | Simultaneous acquisition of parameters.            | <b>7-in-1 Parallel Acquisition</b> of N, P, K, pH, EC, Temp, and Moisture.    |

*All actual results were validated through hardware-in-the-loop testing.*

# Mapping to POs

The Embedded AI-Based Soil Analysis Device (Soil Doctor) aligns with the Program Outcomes (POs) as follows:

| PO  | Correlation | Justification                                                                                                                     |
|-----|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| PO1 | 3           | Applied embedded systems, industrial communication (RS485), and CRC-16 math to solve complex soil nutrient monitoring problems.   |
| PO2 | 3           | Formulated the "Information Lag" problem and used first-principles data modeling to reach substantiated agricultural conclusions. |
| PO3 | 3           | Designed a portable, battery-powered solution for small-scale farmers with consideration for environmental sustainability.        |
| PO4 | 2           | Validated system accuracy through hardware-in-the-loop testing and virtual simulations against TNAU benchmarks.                   |
| PO5 | 3           | Effectively applied modern tools like Arduino IDE, Spyder (Python), EasyEDA, and Wokwi for system development.                    |

# Mapping to POs

| PO   | Correlation | Justification                                                                                                                |
|------|-------------|------------------------------------------------------------------------------------------------------------------------------|
| PO7  | 3           | Promoted sustainable development by minimizing chemical runoff through precision fertilizer application.                     |
| PO9  | 3           | Functioned effectively as a collaborative team (Group 05) with clear roles in circuit design, software, and testing.         |
| PO10 | 3           | Communicated complex engineering details through technical reports, LCD interfaces, and professional presentations.          |
| PO11 | 2           | Managed project milestones and finances through a structured 9-phase timeline for a low-cost architecture.                   |
| PO12 | 3           | Demonstrated life-long learning by exploring Modbus-RTU, hardware signal conversion, and offline machine learning inference. |

# Sustainable Development Goals (SDGs)

- **SDG 2: Zero Hunger :**

The proposed ESP32-based soil monitoring system aids in improving crop productivity by analyzing soil nutrient levels and providing appropriate crop and fertilizer recommendations.

- **SDG 9: Industry, Innovation and Infrastructure:**

The project encourages innovation through the application of embedded systems technology to develop a low-cost, portable, and efficient agricultural monitoring solution.

- **SDG 12: Responsible Consumption and Production :**

By enabling precise fertilizer recommendations, the system helps reduce excessive fertilizer usage and promotes sustainable agricultural practices.

- **SDG 15: Life on Land :**

Accurate soil nutrient monitoring supports soil health preservation and minimizes long-term land degradation.

# Conclusion

## Societal & Technical Relevance

- **Societal:** Empowers farmers with an affordable, real-time alternative to expensive and time-consuming laboratory soil testing.
- **Technical:** Integrates industrial sensing with **Embedded AI** (Decision Tree) for near-instant, offline inference on the ESP32.

## Merits of the Design

- **Industrial Reliability:** RS485 protocol with 16-bit CRC for noise-resistant, error-free field data.
- **7-in-1 Sensor Synergy:** Single-probe acquisition of NPK, pH, EC, Moisture, and Temperature.
- **Explainable Edge AI:** Hard-coded Decision Tree on ESP32 for 95% accurate, offline inference.
- **Agronomic Precision:** Fertilizer advice grounded in official TNAU Agritech benchmarks.
- **Autonomous Portability:** Handheld unit with a custom 5V rechargeable module for wireless use.
- **Efficiency & Privacy:** High-speed C logic ensures instant, secure, and fully offline processing.

# Future Extensions

- **Automated Smart Irrigation:** Integration of IoT and LPWAN technologies (specifically LoRa) to enable automated irrigation management.
- **Mobile & Cloud Synchronization:** Analysis results and crop recommendations will be synced with a cloud-connected mobile application that maintains specific irrigation requirements for different crops.
- **Fixed Field Controller:** Installation of a permanent field controller to continuously monitor soil moisture and automatically regulate pump or sprinkler systems.
- **Long-Range Communication:** Utilization of LoRa-based communication for reliable, low-power data transmission over long distances, ideal for large-scale agricultural environments.
- **Real-Time Monitoring Dashboards:** Implementation of IoT-enabled cloud dashboards to provide live tracking of irrigation status and overall system performance.
- **Data Logging:** Transitioning from an offline device to one integrated with IoT platforms for long-term data logging and remote monitoring.

# References

- [1] S. Paudel, R. Nakarmi, P. Giri, and S. B. Karki, "Prediction of crop yield based on soil moisture using machine learning algorithms," in *Proc. 2nd Int. Conf. on Technological Advancements in Computational Sciences (ICTACS)*, IEEE, 2022, pp. 491–495.
- [2] A. Khaliq, A. Khan, S. Jan, M. Umair, A. Gulshair, A. Ali, and U. A. Shah, "AI-driven smart agriculture: An integrated approach for soil analysis, irrigation, and crop-fertilizer recommendations," *IEEE Access*, 2025.
- [3] Tamil Nadu Agricultural University (TNAU), "Fertilizer Schedule for Vegetable Crops," *Agritech Portal*, 2024. [Online]. Available: [agritech.tnau.ac.in](http://agritech.tnau.ac.in)

# Approval from Guide

Approved

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